

AP CALCULUS AB - UNIT 1

Guided Notes 1.7: Selecting Procedures for Determining Limits



Name: _____ Class: _____ Date: _____

Learning Objectives and Essential Question

Essential question: How can we diagnose a limit before calculating?

- Diagnose a limit before calculating.
- Match algebraic structure to an efficient valid procedure.
- Choose among analytical, graphical, tabular, one-sided, and theorem-based evidence.
- Verify a result through a second representation or local check.

Key Knowledge, Vocabulary, and Notation

Diagnose	Substitute or inspect the representation to identify the form and available evidence.
Select	Choose the shortest method licensed by the structure.
Execute	Carry out the method while preserving domain and one-sided conditions.
Verify	Check sign, scale, graph, table, or a second algebraic form.
Inconclusive is information	A failed law or $0/0$ tells you to change procedures; it is not a final answer.

Concept Development and Strategy

1. Use the cycle: diagnose $\rightarrow$ select $\rightarrow$ execute $\rightarrow$ verify.
2. Do not factor radicals, rationalize polynomials unnecessarily, or force algebra when only a graph/table is provided.
3. For composite and piecewise limits, track the direction of the inner landing value before selecting the outer branch.

Shared Representation / Data

Observed form / evidence	Recommended first move
Finite substitution	Direct evaluation
$0/0$ with polynomial factors	Factor and cancel
$0/0$ with radicals	Conjugate
Trig input $\rightarrow 0$	Create a standard trig anchor
Absolute value / piecewise target	Analyze one side at a time
Only graph or table supplied	Read the available representation

Worked Example: Procedure choice

Choose the first move for $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$.

Answer and Reasoning

Substitution gives 0/0 and the numerator is a difference of squares, so factor and cancel.

Worked Example: Two transformations

Choose a plan for $\lim_{x \rightarrow 0} \frac{\sqrt{1+x}-1}{\sin x}$.

Answer and Reasoning

Rationalize the numerator and create the trig anchor: $\frac{x}{(\sqrt{1+x}-1)\sin x} = \frac{x}{\sin x} \frac{1}{\sqrt{1+x}-1} \rightarrow 1 \cdot \frac{1}{2}$.

Worked Example: Directional composition

If $g(x) \rightarrow 2^+$ and $F(t) = 1$ for $t \leq 2$, $F(t) = t + 4$ for $t > 2$, find $\lim F(g(x))$.

Answer and Reasoning

The inner values approach from above, so the outer right branch applies and the limit is 6.

Guided Practice and Discussion

Directions

Show the mathematical setup, preserve exact values unless a decimal is requested, and justify existence or interpretation statements with the appropriate definition or theorem.

1. Select the best first method for $\lim_{x \rightarrow 2} (3x^2 - 1)$.

2. Select the best first method for $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$.

3. Substitution in a quotient gives $5/0$. What should you inspect next?

4. A problem provides only a graph. Why is “factor the expression” not an available procedure?

5. Choose a complete plan for $\lim_{x \rightarrow 0} \frac{1 - \cos(2x)}{x \sin(3x)}$.

6. A student uses direct substitution on $\frac{x^2 - 1}{x - 1}$ at $x = 1$ and reports DNE. Diagnose the reasoning failure.

7. Design a decision tree with at least four branches for evaluating limits from formulas.

AP Reasoning Check

Multiple-Choice Check

Choose the best answer. Explain why at least one distractor is incorrect during class discussion.

1. After substitution, $\lim_{x \rightarrow c} \frac{f(x)}{g(x)}$ has form $7/0$. The best next step is to
 - (A) cancel 7
 - (B) inspect denominator signs from each side
 - (C) report 0
 - (D) use the product law
2. Which procedure is most appropriate for $\lim_{x \rightarrow 9} \frac{x-9}{\sqrt{x}-3}$?
 - (A) Long division
 - (B) Conjugate rationalization
 - (C) Squeeze theorem
 - (D) Numerical integration
3. A calculator graph and table disagree near $x = 2$ because the graph window is coarse. Which evidence should be refined first?
 - (A) The title of the function
 - (B) The local graph window and table precision
 - (C) The function value at $x = 0$
 - (D) A distant interval

Common Errors and AP Precision

- Keep the value of a function at a point separate from its limiting behavior near that point.
- State one-sided behavior explicitly whenever a two-sided limit or continuity claim depends on agreement from both sides.
- A complete AP justification names the relevant definition, condition, or theorem rather than reporting only a numerical result.

Exit Ticket

Find two valid procedures for $\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$ and explain which is available in a first limits unit.

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